Equation (8) implies that *the optimum HTC is a Bayes detector with costs assigned as functions of the a priori probabilities* (See [2]).

If the receiver has control over the value of p (possibly by influencing transmitter coding), the p to use is that which maximizes $I^*(p)$:

$$I^{**} = I^*(p^*) = \max_p I^*(p). \tag{9}$$

Since $I^*(p)$ was obtained numerically, I^{**} can be obtained analytically only by reversing the order of maximization in (5) and (9):

$$I^{**} = \max_p I^*(p) = \max_p \max_{\alpha,\beta} I(\alpha, \beta, p)$$
$$= \max_{\alpha,\beta} \max_p I(\alpha, \beta, p). \tag{10}$$

However, $\max_p I(\alpha, \beta, p)$ is, by definition, the channel capacity of the HTC. I^{**} is the result of maximizing the channel capacity of the binary asymmetric channel subject to the receiver operating characteristic, $K(\alpha, \beta, R) = 0$:

$$I^{**} = \max_{\alpha,\beta} C(\alpha, \beta) \tag{11}$$

where

$$C(\alpha, \beta) = \max_p I(\alpha, \beta, p)$$
$$C(\alpha, \beta) = I(\alpha, \beta, p^*).$$

$p^*(\alpha, \beta)$ and $C(\alpha, \beta)$ are found in [3].

The maximization indicated in (11) results in (12) which is similar in form to (3):

$$\frac{\partial K}{\partial \alpha}(\alpha, \beta, R) = +\frac{\frac{\partial C}{\partial \alpha}(\alpha, \beta)}{\frac{\partial C}{\partial \beta}(\alpha, \beta)}. \tag{12}$$

Two explicit solutions exist: $\alpha = 1 - \beta$, which is the trivial case, $R = 0$; and $\alpha = \beta$, which is the binary symmetrical channel, BSC,

$$p^*(\alpha = \beta) = \tfrac{1}{2} \tag{13}$$

$$I^{**} = \ln 2 - H\left\{\Phi\left(\frac{R}{2}\right)\right\} \quad \text{(nats)} \tag{14}$$

$$\theta^{**} = \ln 1 = 0. \tag{15}$$

$H(\ )$ is the Shannon entropy function. Equations (13) and (15) imply that *the optimum HTC for the best choice of $p = 1/2$ is the BSC which is the Ideal Observer*. The Ideal Observer is the test that results when the probability of making any error is minimized. Equations (13) through (15) agree with the implications of Table I.

These results may be applied to problems in optical communication. Reiffen and Sherman [4] and Helstrom [5] show that the optimum optical detector compares the logarithm of a likelihood ratio, which is a weighted sum of photon counts, to a threshold. Helstrom comments that when the observation interval is large, the logarithm statistic compared to the threshold approaches a Gaussian random variable by the central limit theorem. Both papers state that, under high background radiation (noise) conditions, the variance of the statistic is the same under both hypotheses (presence or absence of optical signal) and equals the difference between the statistic means under the two hypotheses. Therefore, the optimum optical detector is constrained by (2) with

$$R^2 = \frac{\eta N_s^2}{N_0}$$

where

η = detector quantum efficiency
N_s = average number of signal photons
N_0 = average number of background photons.

In the case of high background radiation, $N_s/N_0 \ll 1$, (14) is applicable. The capacity of a binary photon channel, whose detector threshold has been set to maximize the capacity, is given by (16):

$$C_{\max} = \ln 2 - H\left(\Phi\left\{\frac{N_s}{2}\left(\frac{\eta}{N_0}\right)^{1/2}\right\}\right) \quad \text{(nats)}. \tag{16}$$

Consider what may be done when p is both uncontrolled and unknown. Assume that $I(\alpha, \beta, p)$ is still chosen as the criterion of hypothesis test performance. If the test will be operating repeatedly in time during which p may be assumed constant, a value of p may be assumed initially which can be improved upon by estimates of p made on the basis of later test decisions. This scheme is an adaptive optimum HTC for known p, or an adaptive Bayes detector which changes its cost assignment as better estimates of p are made.

An alternative procedure which presents itself for unknown p is the minimax approach in which the minimum of I over the unknown p is maximized over α and β. This proves fruitless since $\min_p I = 0$ for all α and β. As a last resort one may try to maximize the least upper bound of $I^*(p)$ when p is unknown. This is the same problem that was solved for known and controlled p.

If the exact value of p is unknown, but p is known to be small ($p < 0.01$) or large ($p > 0.99$), then the limiting loci of $p = 0$ or $p = 1$ in Fig. 3 may be used, respectively, to determine the optimum threshold setting to a good approximation.

Incidentally, the meaning of $I(\alpha, \beta, p)$ as an information measure of performance of an HTC when p is unknown is rather vague. The results of choosing more intuitively meaningful information measures may be found by the method of this correspondence to determine if these results are of such a character as to support the choice of their respective measure.

In summary, the procedure outlined above for determining the optimum threshold in a binary hypothesis test consisted of the maximization of an information measure of performance subject to the operating characteristic of the test. Although knowledge and controllability of the a priori distribution affect the order and interpretation of the maximizations, the procedure is general with respect to operating characteristic and information measure.

THOMAS L. GABRIELE
Carlyle Barton Lab.
The Johns Hopkins University
Baltimore, Md.

REFERENCES

[1] W. W. Peterson, T. G. Birdsall, and W. C. Fox, "The theory of signal detectability," *IRE Trans. on Information Theory*, vol. IT-4, pp. 171–212, September 1954.
[2] D. Middleton, *Introduction to Statistical Communication Theory*. New York: McGraw-Hill, 1960.
[3] F. M. Reza, *An Introduction to Information Theory*. New York: McGraw-Hill, 1961.
[4] B. Reiffen and H. Sherman, "An optimum demodulator for Poisson processes: photon source detectors," *Proc. IEEE*, vol. 51, pp. 1316–1320, October 1963.
[5] C. W. Helstrom, "The detection and resolution of optical signals," *IEEE Trans. on Information Theory*, vol. IT-10, pp. 275–287, October 1964.

On the Existence of Perfect Maps

Let A be an $M \times N$ matrix of ones and zeros. It is convenient in what follows to regard A as a *matrix on a torus*, so that its last column is followed by its first column, and its last row is followed by its first row. This can be most easily visualized by placing copies of

Manuscript received November 20, 1965. This research was supported in part by NSF Grant GP-393.

Authorized licensed use limited to: Lakehead University. Downloaded on September 24,2026 at 15:20:43 UTC from IEEE Xplore. Restrictions apply.

A on each of its sides; the process can then be continued until the entire plane is covered with such copies.

Suppose next that $m \leq M$ and $n \leq N$ are positive integers. Then there are 2^{mn} different $m \times n$ matrices of ones and zeros, say $B_1, \cdots, B_{2^{mn}}$. Following Reed and Stewart[1], we call A a *perfect map for an $m \times n$ submap* if

1) each B_i occurs at most once as a submatrix of A whose rows and columns are consecutive in the toroidal sense,
2) $mn = \langle \log_2 MN \rangle$, where $\langle x \rangle$ denotes the least integer $\geq x$.

For example, the matrix

$$A = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 1 \end{bmatrix}$$

is a perfect map for a 2×2 submap. The purpose of this correspondence is to prove the following result.

Theorem

Given positive integers m and n, not both equal to 6, there are integers M, N such that a perfect $M \times N$ map for an $m \times n$ submap exists.

Proof: There is no loss of generality in supposing that $m \geq n$, for the case $m < n$ can then be dealt with by transposing the matrices constructed below. Consider the field $\Omega = GF(2^{mn})$; its nonzero elements Ω^* form a multiplicative group of order $2^{mn} - 1$. Suppose for the moment that $m \neq 6$; the case $m = 6$ will be discussed separately later on. Then by a theorem of A. S. Bang, there is a prime p such that $p \mid 2^m - 1$, but $p \nmid 2^\mu - 1$ for any $\mu < m$ (For an accessible proof see Dickson[2]). Thus m is the order of 2 in the multiplicative group of nonzero residues (mod p). Hence $m \leq p - 1$. Now suppose $p^a \parallel 2^m - 1$ (i.e., $p^a \mid 2^m - 1$ but $p^{a+1} \nmid 2^m - 1$). Then $p \nmid (2^{mn} - 1)/p^a$. For by the definition of a, $p \nmid (2^m - 1)/p^a$. But if $p \mid (2^{mn} - 1)/(2^m - 1)$. Then

$$2^{(n-1)m} + 2^{(n-2)m} + \cdots + 2^m + 1 \equiv 0 \pmod{p}. \qquad (1)$$

Since $2^m \equiv 1 \pmod{p}$, and there are n terms on the left of (1), we get $n \equiv 0 \pmod{p}$, so that $n \geq p \geq m + 1$, a contradiction. Therefore, $(p^a, (2^{mn} - 1)/p^a) = 1$. Put $M = p^a$, $N = (2^{mn} - 1)/p^a$, and let α, β be elements of Ω^* of orders M, N, respectively. Since $(M, N) = 1$, α and β generate Ω^*. Since $p^a \mid 2^m - 1$, α lies in the subfield $K = GF(2^m)$. Since $p \nmid 2^\mu - 1$ for any $\mu < m$, α generates K over $\Pi = GF(2)$, i.e., $K = \Pi(\alpha)$. Now β generates Ω over K, i.e., $\Omega = K(\beta)$ for α and β together generate the group Ω^*.

Since $[\Omega : K] = n$, it follows from these considerations that the elements $\alpha^i\beta^j$, where $0 \leq i < m$, $0 \leq j < n$ form a basis of Ω over Π. Next let L be a linear functional from Ω to Π, not identically 0. Define the $M \times N$ matrix $A = (C_{ij})$ by putting $C_{ij} = L(\alpha^i\beta^j)$, where $0 \leq i < M$, $0 \leq j < N$. Then A is the desired perfect map for an $m \times n$ submap. Indeed, since $\alpha^M = \beta^N = 1$, the extension of A to the entire plane as described in the introduction is simply the matrix $L(\alpha^i\beta^j)$, where $-\infty < i, j < \infty$. Therefore, the $m \times n$ submatrix $A_{rs} = (C_{ij})$, $r \leq i < r + m$, $s \leq j < s + n$ has entries $L(\alpha^r\beta^s\alpha^{i-r}\beta^{j-s})$. Similarly, the submatrix $A_{tu} = (C_{ij})$, $t \leq i < t + m$, $u \leq j < u + n$ has entries $L(\alpha^t\beta^u\alpha^{i-t}\beta^{j-u})$. Thus if $A_{rs} = A_{tu}$, then $L(\alpha^r\beta^s\alpha^k\beta^l) = L(\alpha^t\beta^u\alpha^k\beta^l)$ for all k, l with $0 \leq k < m$, $0 \leq l < n$. Since the elements $\alpha^k\beta^l$ are a basis of Ω over Π, it follows that $L(\alpha^r\beta^s\theta) = L(\alpha^t\beta^u\theta)$ for all $\theta \in \Omega$. But if L' is any linear functional from Ω to Π, then there is a $\theta \in \Omega$ such that $L'(\xi) = L(\theta\xi)$ for all ξ (this is because the number of functionals L' is 2^{mn}, the number of θ's is also 2^{mn}, and the mappings $L(\theta\xi)$ are clearly linear in ξ and distinct for distinct θ's). Hence $L'(\alpha^r\beta^s) = L'(\alpha^t\beta^u)$ for all L', which implies $\alpha^r\beta^s = \alpha^t\beta^u$. Hence $r \equiv t \pmod{M}$ and $s \equiv u \pmod{N}$. This shows that condition 1) in the definition of a perfect map is satisfied. Condition 2) is trivial, since $MN = 2^{mn} - 1$. This completes the proof for the case $m \neq 6$.

It will be observed that all that was required to make the above proof work was the existence of elements $\alpha, \beta \in \Omega^*$ of orders M, N, respectively, where $2^{mn} - 1 = MN$, $(M, N) = 1$, such that the elements $\alpha^i\beta^j (0 \leq i < m, 0 \leq j < n)$ are linearly independent over Π. We will now show the existence of such elements when $m = 6$, $n < 6$, thus completing the proof of the theorem. Choose M and N as in the following table:

m	n	M	N
6	1	$3^2 \cdot 7$	1
6	2	$3^2 \cdot 7$	$5 \cdot 13$
6	3	$3^3 \cdot 19 \cdot 73$	7
6	4	$3^2 \cdot 7$	$5 \cdot 13 \cdot 17 \cdot 241$
6	5	$3^2 \cdot 7$	$11 \cdot 31 \cdot 151 \cdot 331$

Let α and β be any elements of Ω^* of orders M and N, respectively. When $n = 1$, 2, 4, or 5, α is a primitive element of $K = GF(2^m)$ over Π, and the proof proceeds as before. If $n = 3$, β is a generator of $\Lambda = GF(2^n)$ over Π, and α is a generator of Ω over Λ. Hence $\alpha^i\beta^j (0 \leq i < 6, 0 \leq j < 3)$ are linearly independent over Π. This concludes the proof of the theorem.

It seems very likely that the same conclusion holds for $m = n = 6$. In this case $2^{mn} - 1 = 3^3 \cdot 5 \cdot 7 \cdot 13 \cdot 19 \cdot 37 \cdot 73 \cdot 109$. For the above construction to work, we must choose integers M and N with $(M, N) = 1$, $MN = 2^{mn} - 1$, and then find elements $\alpha, \beta \in \Omega$ of orders M, N such that the elements $\alpha^i\beta^j (0 \leq i, j < 6)$ are linearly independent over Π. Taking $M > N > 6$, there are 126 possible choices for M and N. For each such choice we can take $\alpha = \gamma^N$, $\beta = \gamma^M$, where γ is a generator of the group Ω^*. For example, we can choose $\gamma = x^5 + x + 1$, where x satisfies the equation $x^{36} + x^{35} + \cdots + x + 1 = 0$ over Π. Then the powers $\alpha^i\beta^j$ can be expressed in the form $C_0 + C_1x + \cdots + C_{35}x^{35}$, and the question is reduced to showing that the 36×36 determinant whose rows are the vectors $(C_0, C_1, \cdots, C_{35})$ is not zero. We have not carried out the detailed computations.[3]

In conclusion we note that the perfect maps A constructed above have the property that every $m \times n$ matrix (of ones and zeros) except the zero matrix occurs in A. Some authors require that the zero matrix must also occur in A; then we clearly have $MN = 2^{mn}$. The Reed-Stewart definition used in this correspondence appears to be more flexible, and to lead to a richer theory.

BASIL GORDON
Dept. of Mathematics
University of California
Los Angeles, Calif.

[1] I. S. Reed and R. M. Stewart, "Note on the existence of perfect maps," *IRE Trans. on Information Theory*, vol. 8, pp. 10–12, January 1962.
[2] L. E. Dickson, "On the cyclotomic function," *Am. Math. Monthly*, vol. 12, pp. 86–89, 1905.
[3] Added in proof: These computations have since been made by Christopher Landauer, a student at the University of California at Los Angeles. As suspected, there are indeed perfect maps for a 6×6 submap, so the above theorem has no exceptions.

Authorized licensed use limited to: Lakehead University. Downloaded on September 24,2026 at 15:20:43 UTC from IEEE Xplore. Restrictions apply.